GRNsight v2:

a web application for visualizing models of small gene regulatory networks

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SWE Rapid Talk - Undergraduate

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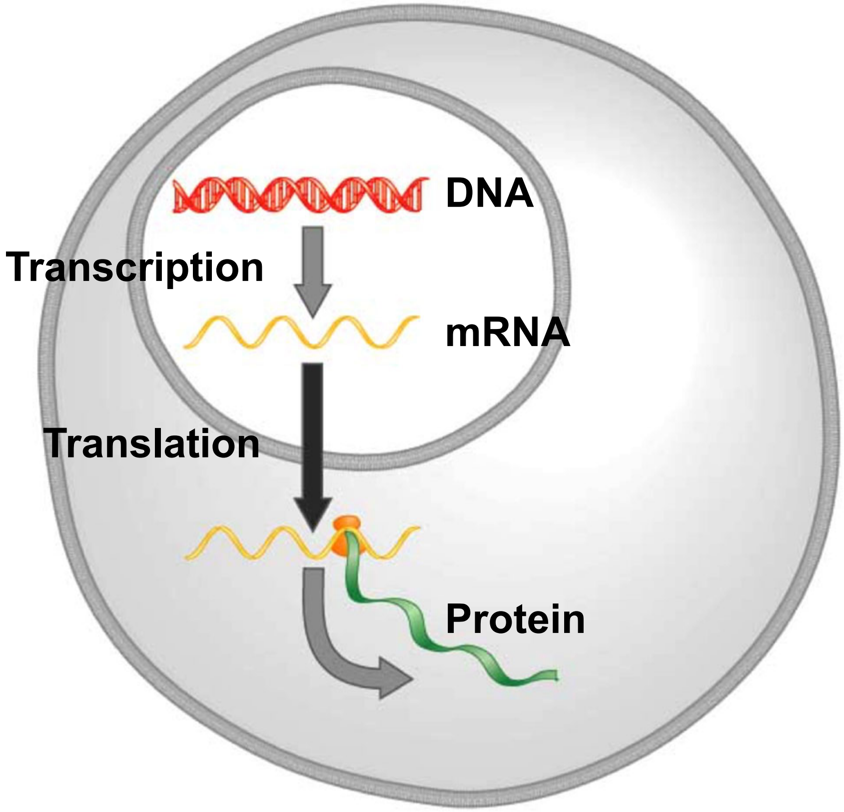
Frank R. Seaver College
of Science and Engineering

A gene regulatory network (GRN) controls the level of expression of genes and proteins



Gene regulatory process

- Transcription factors **activate** or **repress** expression.
- To understand GRNs we want to quantify the **weight values** of these regulatory relationships

GRNmap models the dynamics of GRNs to estimate weight parameters (*w*)

MATLAB

- Magnitude of weights represent the influence of the transcription factor on its target gene
- **Positive** weights = **activation** of gene expression.
- **Negative** weights = **repression** of gene expression.

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1+\exp\left(-\left(\sum_j \left(w_{ij}x_j(t)\right)-b_i\right)\right)} - d_i x_i(t)$$

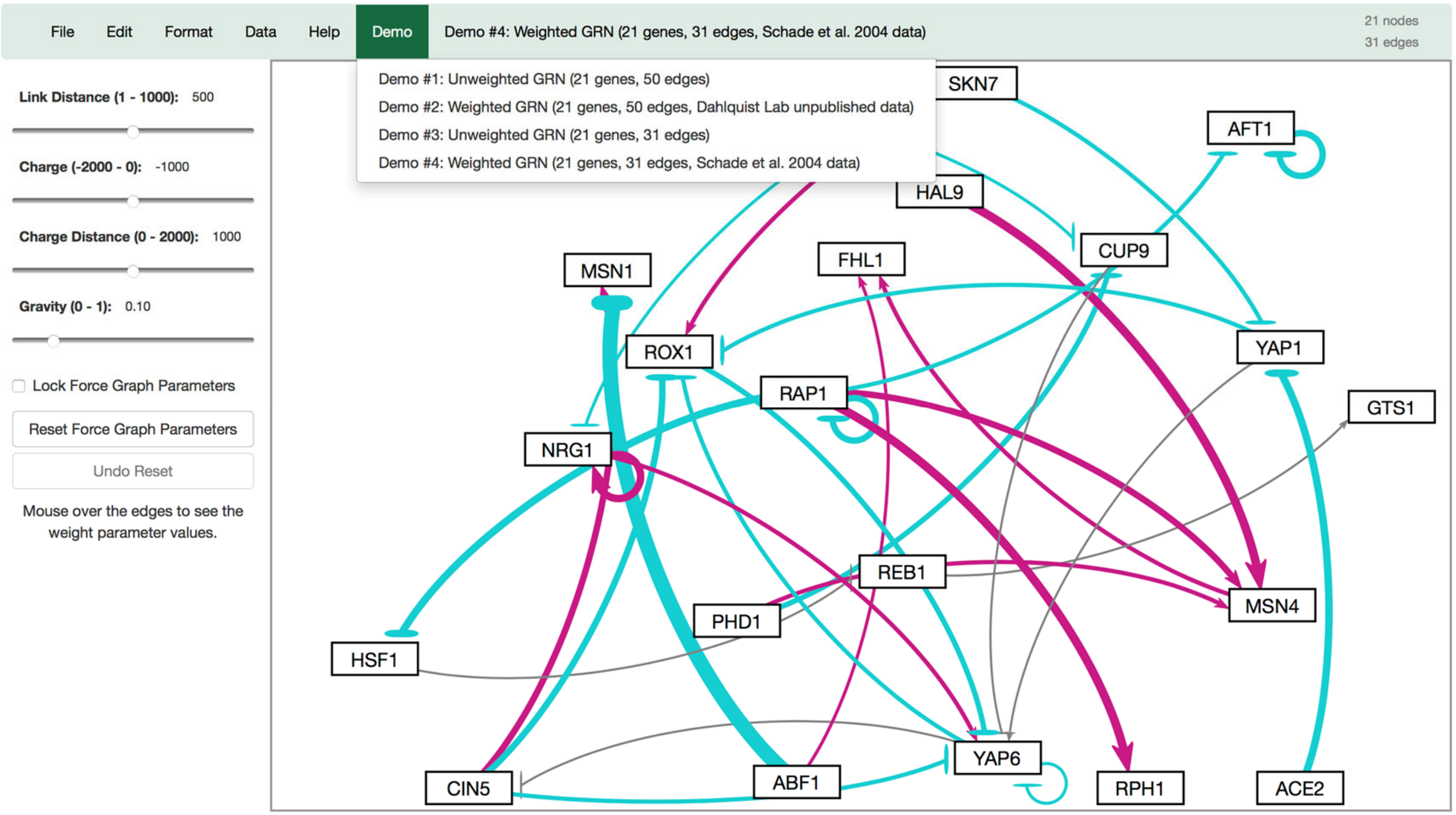
GRNmap differential equation

GRNmap outputs Excel spreadsheets, with estimated weight parameters

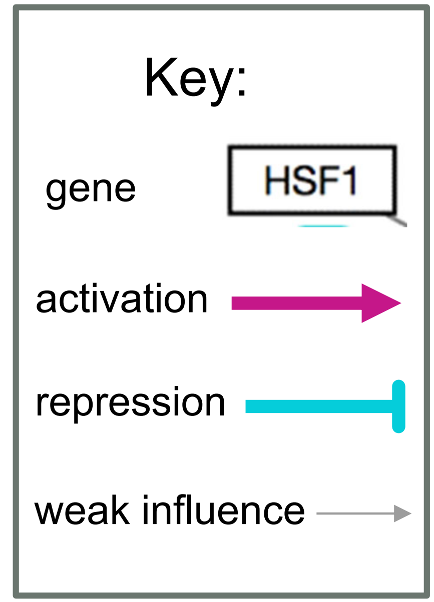
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	cols regulators/rows targets	ABF1	ACE2	AFT1	CIN5	CUP9	FHL1	GTS1	HAL9	HSF1	MAC1	MSN1	MSN4	NRG1	PHD1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP6
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFT1	0	0	-0.8966	0	0	0	0	0	0	0	0	0	0	0	-0.4030	0	0	0	0	0	0
5	CIN5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.0450
6	CUP9	0	0	0	0	0	0	0	0	0	-0.1882	0	0	0	-0.6510	0	0	0	0	0	0	0
7	FHL1	0.1562	0	0	0	0	0	0	0	0	0	0	0.6121	0	0	0	0	0	0	0	0	0
8	GTS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0778	0	0	0	0	0
9	HAL9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.2321	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.9393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	1.4283	0	0	0	0	0	0.5447	1.0131	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	1.2341	0	0	0	0	0	-0.1852	0	0
15	PHD1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.8890	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	-0.0102	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9278	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	-0.4315	-0.5071
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.4999	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3615	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4082	0	0
22	YAP6	0	0	0	-0.5312	-0.1293	0	0	0	0	0	0	0	0.6215	0	0	0	-0.7503	0	0	0.0146	-0.3027

21 gene, 31 edge weighted network output from GRNmap

GRNsight automatically produces visualizations of weighted networks from adjacency matrix inputs



Screenshot of GRNsight v1, released Aug 2016

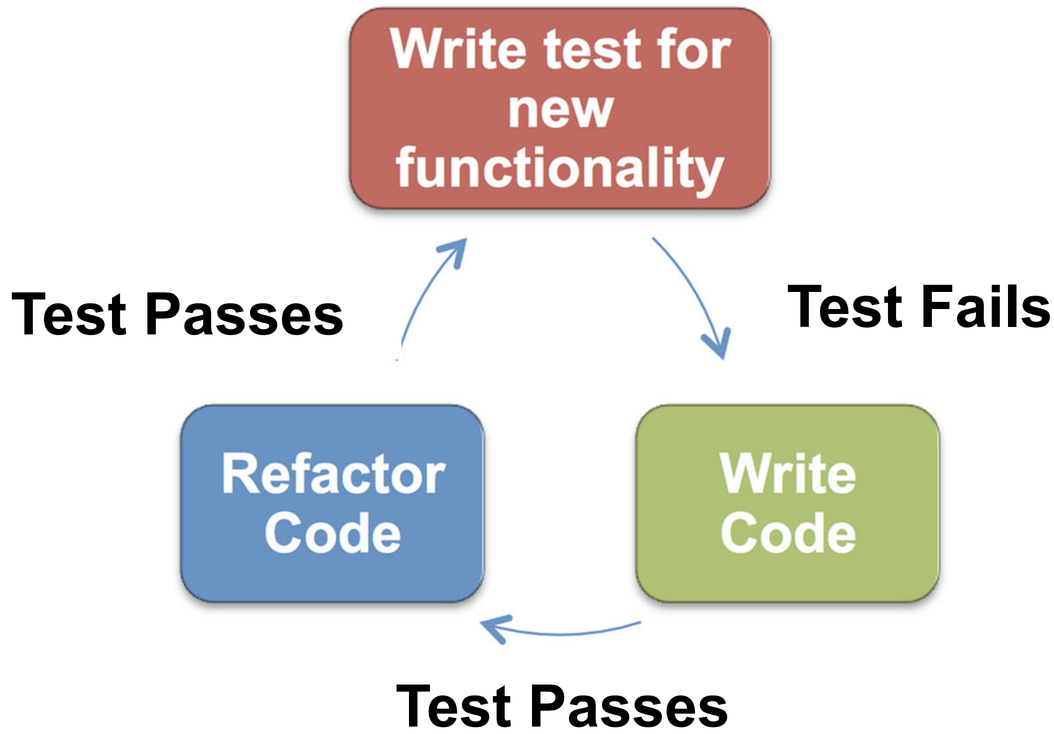


- ✓ **Exists as a web application**
- ✓ **Optimized for small- to medium-scale graphs**
- ✓ **Automatically lays out weighted network graphs**
- ✓ **Easy to use, and easily interpretable visualizations**

Our objective is to develop GRNsight v2, with expanded visual & analytic features

GRNsight is written in JavaScript and follows software engineering best practices

GRNsight technology stack

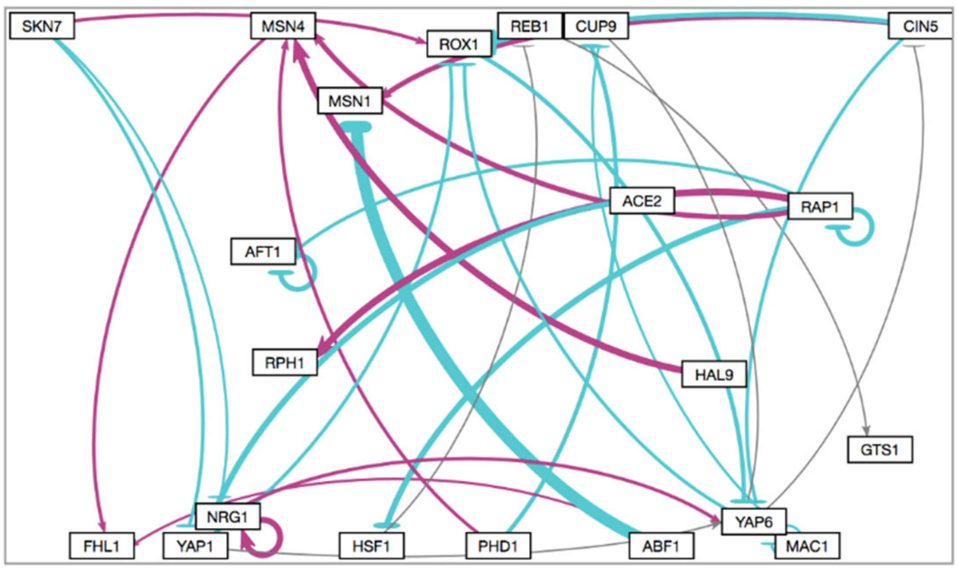


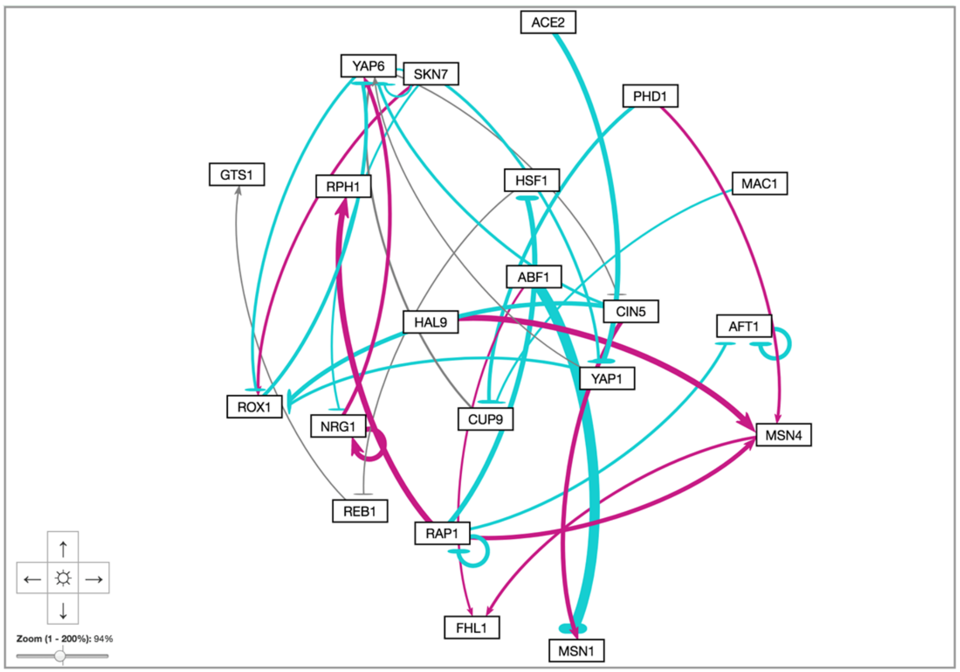
Test driven development cycle

Allow graph to reach steady state

GRNsight v1

GRNsight v2





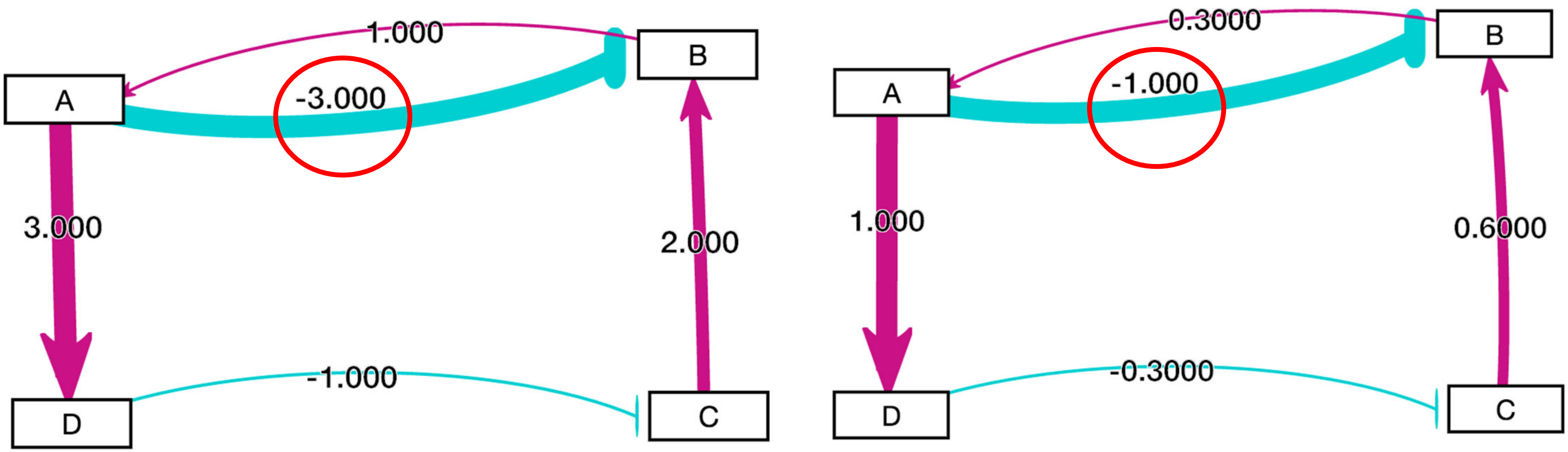
- **Nodes are restricted to the available screen area**
- **No ability to zoom and scroll**

- **Graph bounding box separated from viewport**
- **Zoom and scrolling enabled**

Customizable normalization factor for edge thickness

GRNsight v1

- Visually comparing two graphs was difficult.
- Normalization factor was automatically selected by maximum weight.



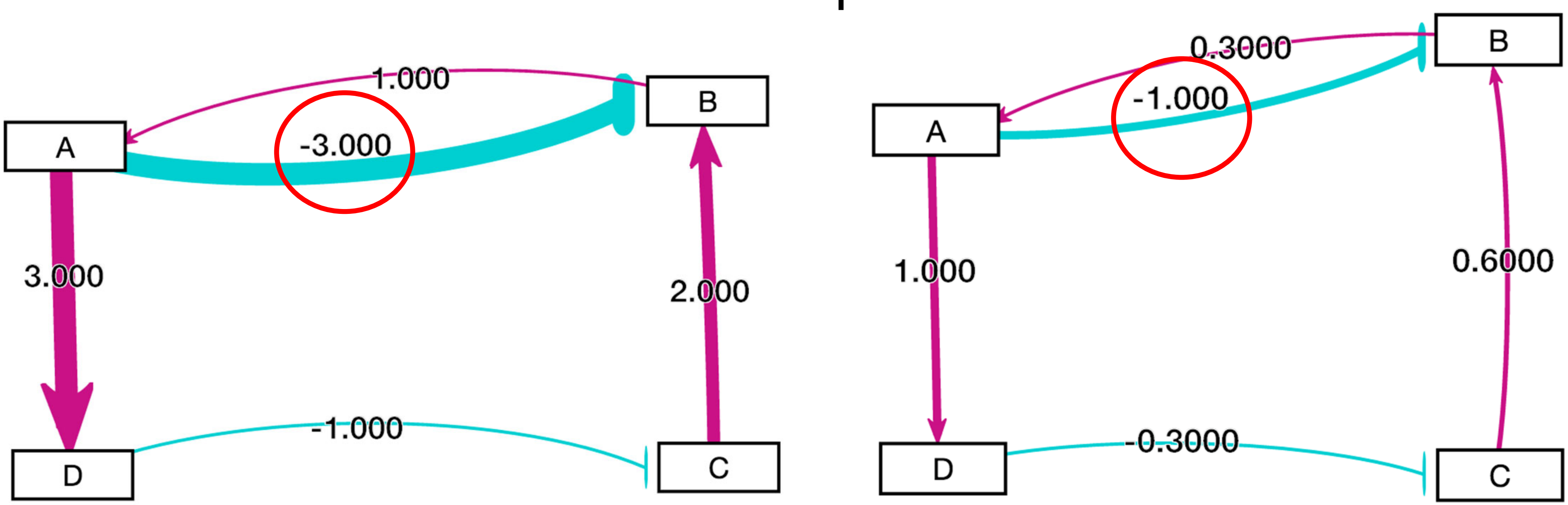
Circled edge thicknesses are the same although weight values are different

Customizable normalization factor for edge thickness

GRNsight v2

- Allows user to set normalization factor using user interface.
- Edge thicknesses can be rendered on the same scale.
- Facilitates accurate visual comparison.

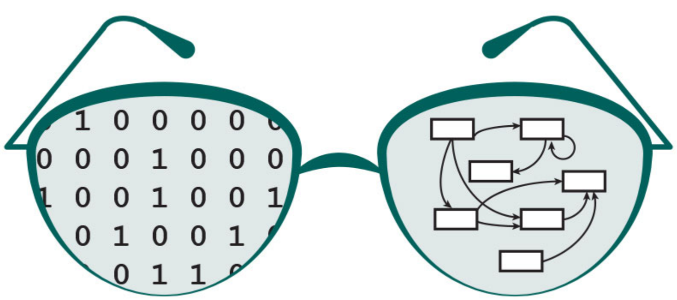
Normalization factor User Interface



Circled edge thicknesses are different when weight values are different

GRNsight v2 improves the interpretation of models of gene regulatory networks

- GRNsight has general applicability for visualizing network graphs across different domains.
- Future work: color the nodes to visualize temporal gene expression data
- GRNsight application and code are available under the open source BSD license

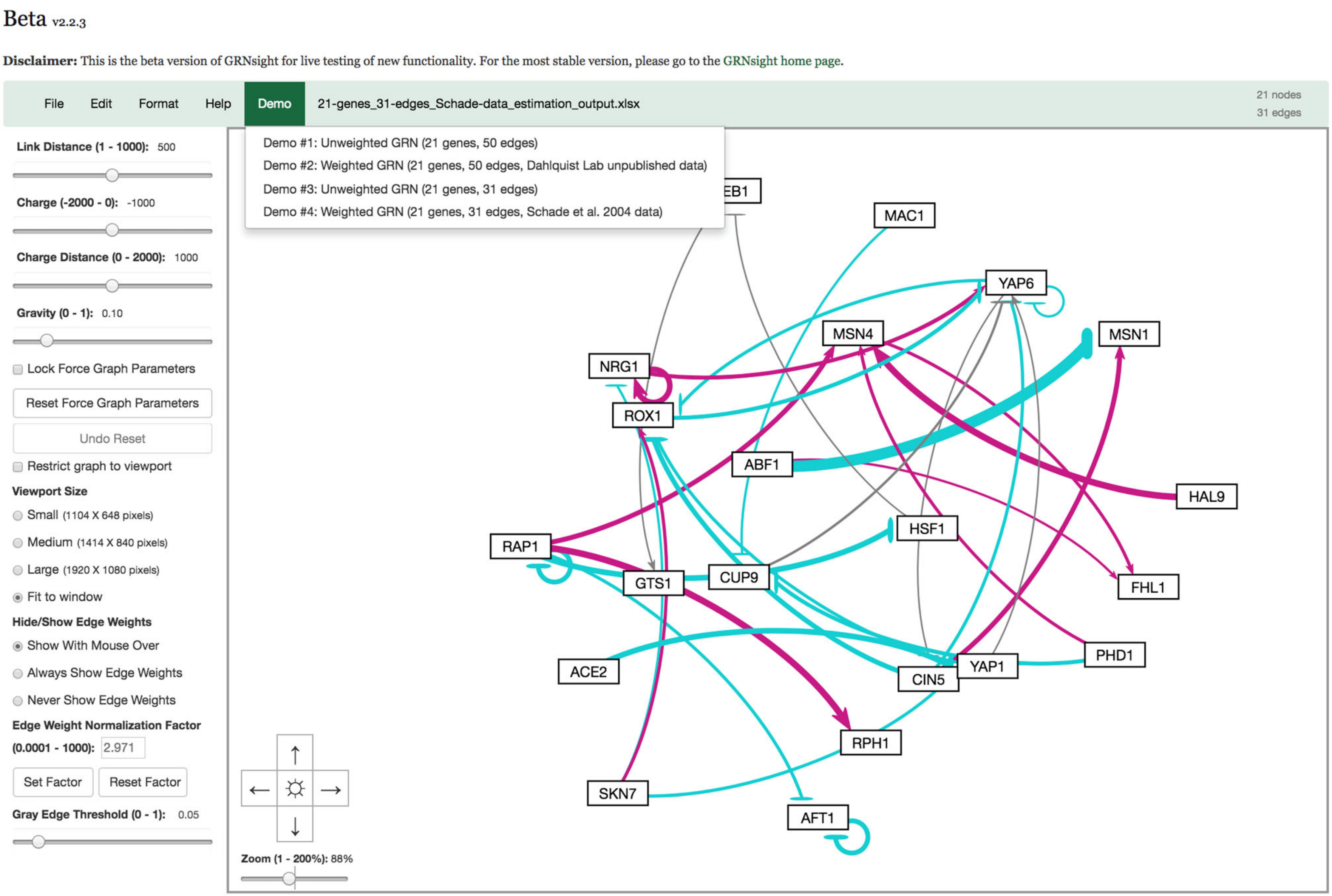


http://dondi.github.io/GRNsight/

https://github.com/dondi/GRNsight

Backup Slides

GRNsight v2.2.3 (beta) Screenshot



GRNsight v2.1.11 (master) Screenshot